Bayesian Betting Odds Analysis of Premier League Relegation Battles

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Abstract

With the rise of sports betting and literature indicating betting odds' usefulness in match prediction, their statistical incorporation has become commonplace in sports analysis. The Premier League relegation battle has long been studied via financial and club structure metrics, but betting odds have traditionally been used for match-level prediction rather than club-level analysis. Using a dataset spanning 20+ years of Premier League matches and betting odds, we propose a Bayesian hierarchical model with odds-to-distribution transformations to examine patterns among teams that survived relegation and those relegated. Our approach converts bookmaker odds into probability distributions, capturing market uncertainty beyond point estimates.We expect this analysis to reveal whether survivors exhibit distinct patterns in odds variance, home/away performance splits, and seasonal trajectories compared to relegated teams. Hierarchical shrinkage will address limited relegated-sample sizes while preserving team-specific patterns.

Keywords: hierarchical modeling, odds to distribution transformation, shrinkage

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1 Introduction

Body of paper. Margins in this document are roughly 0.75 inches all around, letter size paper.

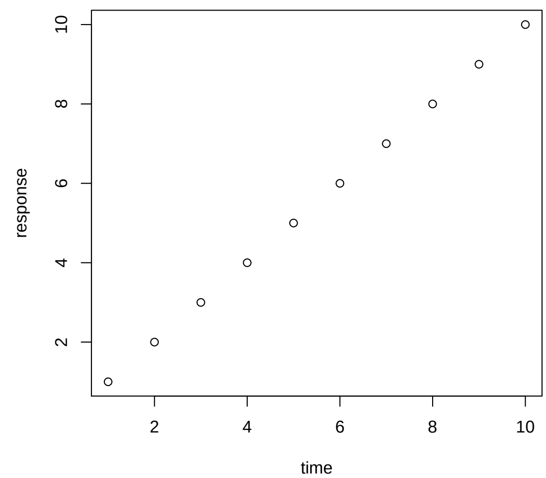

Figure 1: Consistency comparison in fitting surrogate model in the tidal power example.

Table 1: D-optimality values for design X under five different scenarios.

one	two	three	four	five
1.23	3.45	5.00	1.21	3.41
1.23	3.45	5.00	1.21	3.42
1.23	3.45	5.00	1.21	3.43

- Note that figures and tables (such as Figure 1 and Table 1) should appear in the paper, not at the end or in separate files.
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in the text. That is, don't say "In Smith et. al. (2009) we showed that …". Instead, say "Smith et. al. (2009) showed that …".

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- If you have supplementary material (e.g., software, data, technical proofs), identify them in the section below. In early stages of the submission process, you may be unsure what to include as supplementary material. Don't worry—this is something that can be worked out at later stages.

2 Methods

Don't take any of these section titles seriously. They're just for illustration.

3 Verifications

This section will be just long enough to illustrate what a full page of text looks like, for margins and spacing.

Gelman & Vehtari (2021) offer some guidance about key ideas about statistical ideas. On an unrelated note, spreadsheets are important to use correctly (Broman & Woo 2018). Log-linear models are an attractive way to model categorical data (Bishop et al. 1975).

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4 Conclusion

5 Disclosure statement

The authors have the following conflicts of interest to declare (or replace with a statement that no conflicts of interest exist).

6 Data Availability Statement

Deidentified data have been made available at the following URL: XX.

SUPPLEMENTARY MATERIAL

Title: Brief description. (file type)

R-package for MYNEW routine: R-package MYNEW containing code to perform the diagnostic methods described in the article. The package also contains all datasets used as examples in the article. (GNU zipped tar file)

HIV data set: Data set used in the illustration of MYNEW method in Section 3 (.txt file).

7 BibTeX

We encourage you to use BibTeX. If you have, please feel free to use the package natbib with any bibliography style you're comfortable with. The .bst file agsm has been included here for your convenience.

References

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